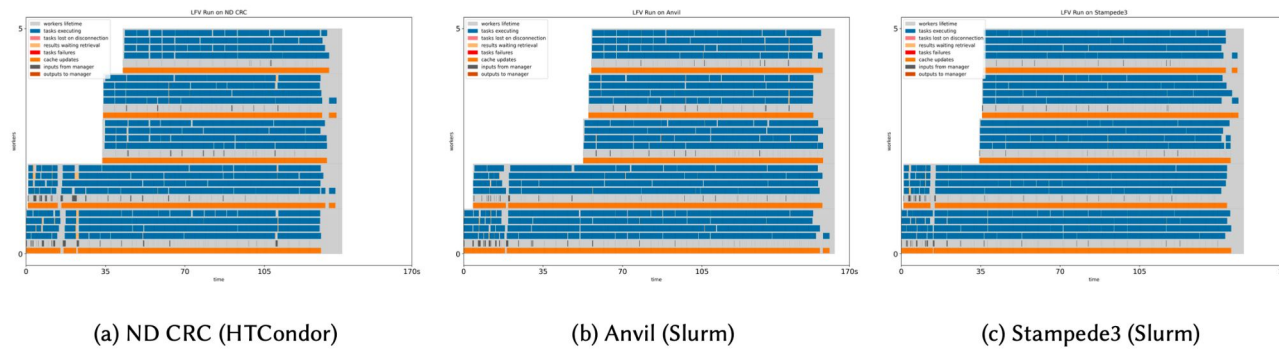


Portable Execution of Data-Intensive Notebook Workflows Across Heterogeneous HPC Systems

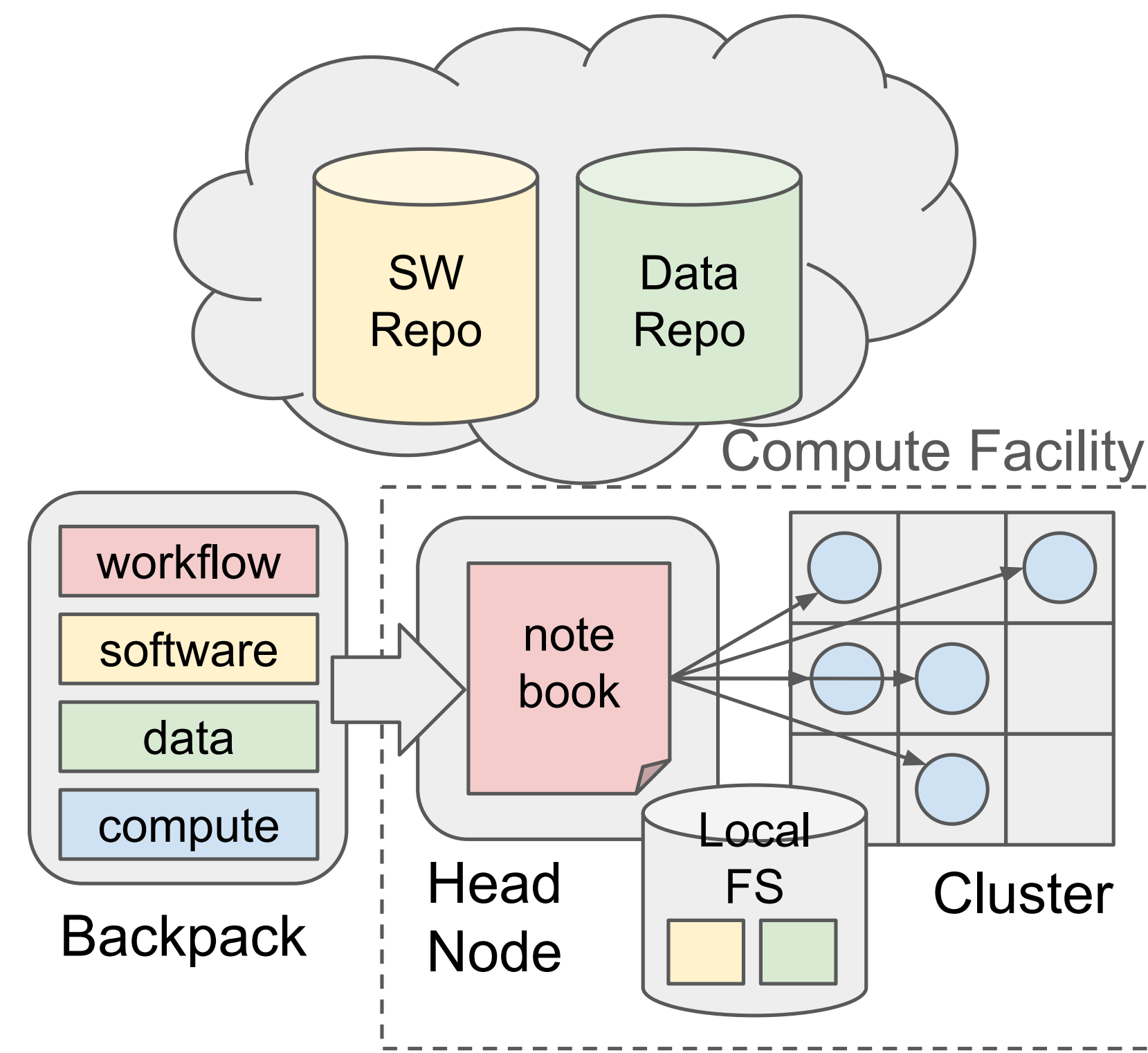
Md Saiful Islam and Douglas Thain
{mislam5, dthain} @nd.edu



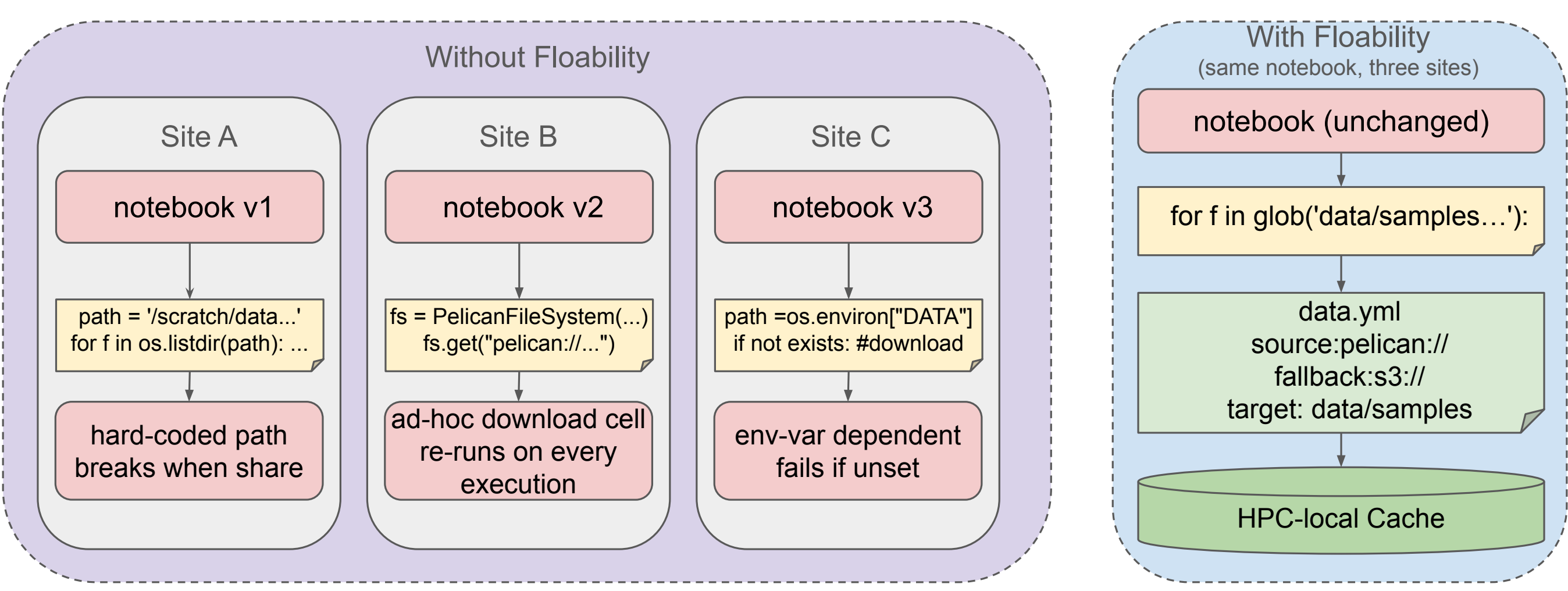
Background and Motivation

- Notebook workflows don't reproduce across HPC sites
- Data is the dominant portability barrier, not code
- Hard-coded paths and ad-hoc downloads break on every move
- **Backpack** packages code, software, data, and resources together
- **Declarative data spec** separates data from notebook code
- **HPC-local cache** makes repeat runs fast

Backpack Abstraction



Portability Gap

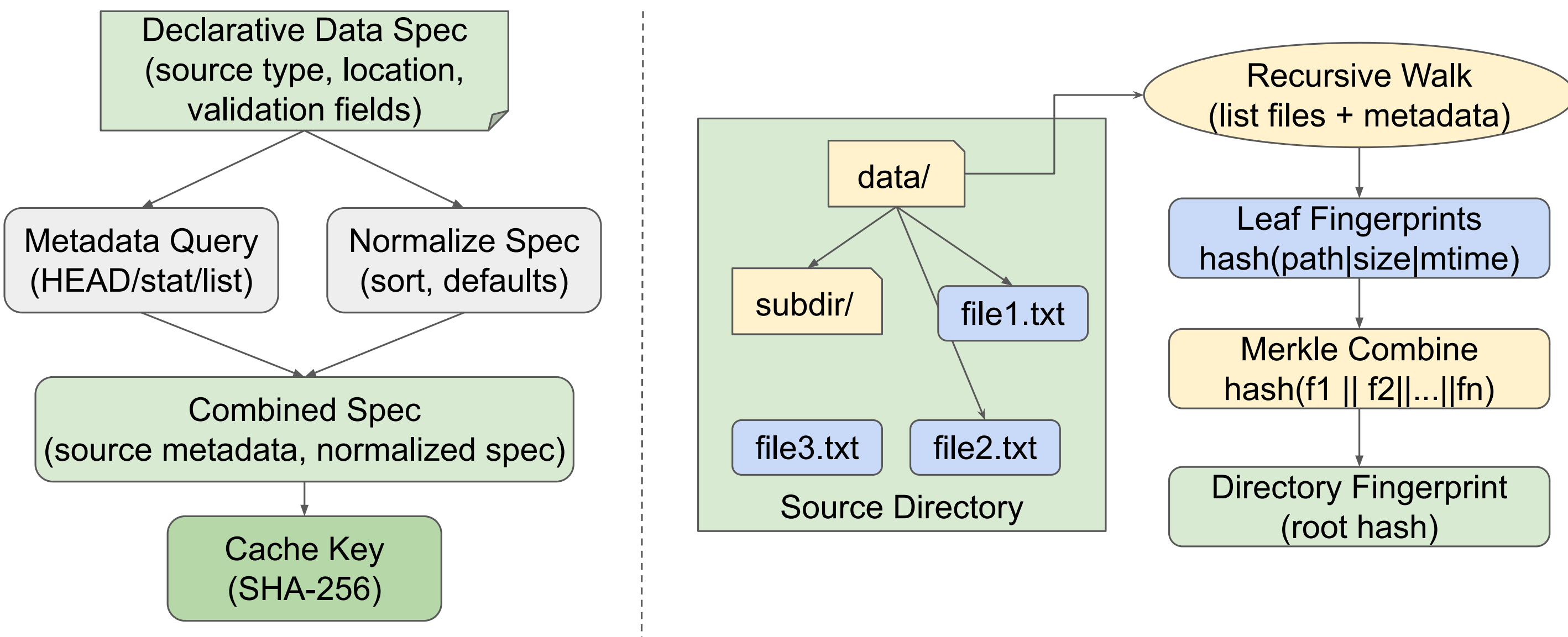


Declarative Data Specification

```
profiles:
  train_set:
    policy:
      retry_attempts: 0
      timeout: 30
    data:
      - name: train_data
        source_type: pelican
        source: pelican://example-data-service.org
          :443/project/datasets/train/
        target_path: data/samples

  test_set:
    data:
      - name: test_data
        source_type: pelican
        source: pelican://example-data-service.org
          :443/project/datasets/test/
        expected_size: 41943040
        checksum: sha256:cccccccccccccc...
        target_path: data/samples
```

Data Fingerprint Generation



Data Operations

